Name :________________________

ChE 311 Midterm Exam I, Spring 2019
Closed book and closed notes. Only provided tables are allowed.
50 minutes, 100 points

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1. (5 points) For the following velocity profile,

$$\vec{v} = (5t + 7x - 3y + z)\overrightarrow{\delta_x} + (x - 2y + z)\overrightarrow{\delta_y} + (5t + x - y + 5z)\overrightarrow{\delta_z}$$

The flow is incompressible.

(A) True

(B) False

(C) Not enough information to decide

2. (5 points) The flow defined by the following velocity profile with a cylindrical coordinate satisfies the continuity equation. For the flow, density is constant.

$$\vec{v} = (-5r)\overrightarrow{\delta_r} + (2t + r + 5z)\overrightarrow{\delta_\theta} + (5z)\overrightarrow{\delta_z}$$

(A) True

(B) False

(C) Not enough information to decide

3. (10 points) Please choose the answer for the following questions.

$$\mathbf{u} = (3x_1 + 4x_2)\boldsymbol{\delta}_1 + (x_2 + 4x_3)\boldsymbol{\delta}_2 + (x_1 + 2x_3)\boldsymbol{\delta}_3$$

(1) $\nabla \cdot \mathbf{u} =$

(a) 16 (b) 6 (c) 10 (d) 5

(2) $\nabla \mathbf{u} =$

(a) $\begin{vmatrix} 3 & 4 & 1 \\ 1 & 1 & 4 \\ 1 & 1 & 2 \end{vmatrix}$ (b) $\begin{vmatrix} 3 & 4 & 0 \\ 0 & 1 & 4 \\ 1 & 0 & 2 \end{vmatrix}$

(c) $\begin{vmatrix} 3 & 1 & 1 \\ 4 & 1 & 1 \\ 1 & 4 & 2 \end{vmatrix}$ (d) $\begin{vmatrix} 3 & 0 & 1 \\ 4 & 1 & 0 \\ 0 & 4 & 2 \end{vmatrix}$

4. (5 points) $\boldsymbol{\tau} = \begin{vmatrix} 2x_1 + x_2 & x_2 + x_3 & 3x_1 + x_3 \\ x_1 & x_2 & x_3 \\ x_3 & x_2 & x_1 \end{vmatrix}$

Please choose the right answer for $\nabla \cdot \boldsymbol{\tau}$

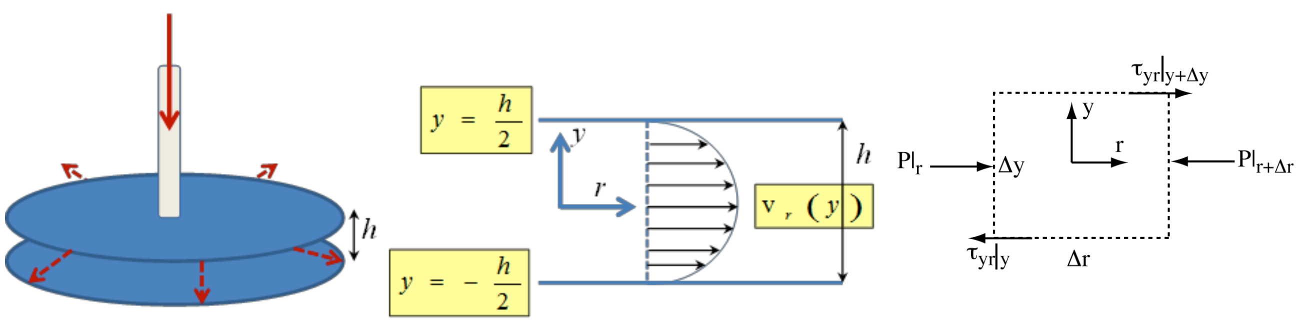
(a) $\nabla \cdot \boldsymbol{\tau} = \begin{vmatrix} 2 & 1 & 1 \\ 1 & 1 & 4 \\ 0 & 1 & 0 \end{vmatrix}$ **(b)** $\nabla \cdot \boldsymbol{\tau} = \begin{vmatrix} 2 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 4 & 0 \end{vmatrix}$

(c) $\nabla \cdot \boldsymbol{\tau} = 3\boldsymbol{\delta}_1 + \boldsymbol{\delta}_2 + 3\boldsymbol{\delta}_3$ **(d)** $\nabla \cdot \boldsymbol{\tau} = 4\boldsymbol{\delta}_1 + 3\boldsymbol{\delta}_2 + \boldsymbol{\delta}_3$

5. A radial-flow channel has been developed to measure the adhesion (energy) of cells to surfaces *in vitro*. The radial-flow channel is shown in the figure below. In this device, fluid flows in from a central source, flows out in a radial direction and the volumetric flow rate at any positions r is Q. The flow itself is driven by an unspecified pressure gradient.

(1) Solve for the velocity profile (20 points).

(2) Use your solution for the velocity profile and the constant flow rate condition to calculate the wall shear stress at r = 1 cm. The fluid viscosity is similar as water 1 mPa•s. The volumetric flow rate is 10 μL/min, with h = 200 μm. (20 points)



6. You need to design a piece of pipe flow for a microfluidic system. Consider the flow is at steady state. Neglect the "end effects". The liquid is a Newtonian fluid with constant density and viscosity. $\nu = 2\times10^{-4}\,m^2/s$ is the kinematic viscosity of the fluid. The density of the fluid is $\rho = 0.8\times10^3\,kg/m^3$. The circular pipe is placed horizontal (figure 1). The length of the pipe is 1.5 cm. Flow is from left to right and driven by a syringe pump. The pressure drop for this piece of pipe is P_1-P_2= 0.25 N/m^2. **(hint: P=P(z)).** The special material used to make this pipe can only afford the maximum shear stress 0.02 N/m^2 before yielding.

(1) With above yield stress, what is the maximum radius of the tube, R? (25 points)
(2) What is the maximum velocity? (10 points)